2025

PREDICTIVE ANALYSIS AND RISK EVALUATION OF DMART STOCK PRICES

From Price Movements to Risk Sensitivity: A Comprehensive Study on DMart

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PROJECT REPORT ON

Predictive Analysis and Risk Evaluation of DMart Stock Prices

From Price Movements to Risk Sensitivity: A Comprehensive Study on DMart

A PROJECT SUBMITTED TO

Bhavan's College

An Academic Project Report Submitted in Partial Fulfilment of the Course

Master of Science in Applied Statistics, Part -1

SEMESTER I

BY

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UNDER THE GUIDANCE OF

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DEPARTMENT OF STATISTICS

(2025-26)

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Acknowledgement

I sincerely thank **Prof. RUHEEN QURESHI** for their expert guidance, valuable feedback, and continuous support throughout this project. Their insights were instrumental in shaping the methodology and ensuring academic rigor.

I am grateful to the **Department of Statistics, Bhavan's College**, for providing an enriching learning environment and access to the necessary resources that made this analysis possible.

Special thanks to **Yahoo Finance** and the **National Stock Exchange (NSE)** for providing the stock price data used in this study, which formed the foundation of all analyses and forecasts. I also appreciate the constructive discussions with peers, which helped refine the clarity and depth of this work.

Finally, I would like to highlight the development of **two Python-based applications**: one for **DMart's next-day price prediction**, and another general-purpose app capable of **predicting the next-day price from any uploaded CSV file**. These tools showcase the practical implementation of the methodologies explored in this project.

"This project demonstrates how statistical learning transforms raw market data into actionable investment insights."

Finally, I thank my family and friends for their encouragement and patience, which kept me motivated throughout the research and writing process.

Shaikh Maryam

21 Sep 2025

1.Abstract:

This project analyzes the stock performance of Avenue Supermarts Ltd. (DMart) to evaluate its investment potential using statistical and machine learning techniques. Historical stock price data from the past two years was examined, focusing on risk assessment, profitability, volatility, and prediction accuracy. The study employed methods such as Markov Chains, Monte Carlo simulations, and time series forecasting to model price behavior and assess investment strategies. Key objectives included determining whether DMart is a risky or safe investment, identifying optimal entry and exit points, and comparing long-term versus short-term investment prospects. Findings suggest that DMart demonstrates moderate volatility with relatively stable returns, indicating a safer investment profile compared to high-beta stocks. The project also highlights the limitations of relying solely on closing prices, while acknowledging that Adjusted Close would improve accuracy. Overall, the analysis provides actionable insights for investors and demonstrates the effectiveness of statistical methods in financial market evaluation.

2.Executive Summary

This project presents a comprehensive analysis of **DMart's stock performance** in comparison with the **NIFTY 50 index**, combining classical statistical approaches with modern machine learning techniques. The primary aim was to assess DMart's risk–return profile, identify patterns in market behavior, and develop predictive tools for practical investment decision-making.

Key Highlights:

- **Statistical Analysis:** Daily returns of DMart were found to be more volatile than NIFTY, with an annualized volatility of **27.06%** vs. **12.55%**, indicating higher risk but also higher opportunity.
- **Risk Assessment:** Using volatility, quartile analysis, and moving averages (20, 50, 200 days), DMart was classified as **"Beneficial"** overall, though subject to short-term bearish phases due to rising competition from online grocery apps.
- **Advanced Models:**
  - **Markov Chain (3-state):** Showed probabilities of return transitions between Up, Down, and Neutral states.
  - **Hidden Markov Model (2-state):** Identified Bull and Bear regimes, revealing hidden cycles in DMart's performance.
  - **Monte Carlo Simulation:** Forecasted a **30-day price range** with a 5–95% confidence interval, expected mean ~₹4850.
- **Risk Metrics:** Value-at-Risk (95%) at **–2.33%**, Conditional VaR at **–2.7%**, and Sharpe Ratio of **–1.6**, reflecting unfavorable risk-adjusted returns compared to the benchmark.
- **Practical Contribution:** Two Streamlit-based applications were developed:

1. **DMart Price Predictor** – predicts tomorrow's stock price with sensitivity and risk analysis.
2. **Universal CSV Predictor** – general tool for any stock dataset.

Conclusion:
DMart offers long-term potential as a defensive retail stock, though it faces short-term unpredictability due to external competition. The integration of statistical tests, regime-switching models, and simulation-based forecasting demonstrates the importance of **data-driven strategies** for informed investment decisions.

3. Introduction:

Stock markets are inherently volatile and influenced by numerous factors such as market sentiment, macroeconomic conditions, and company performance. For investors, the challenge lies in distinguishing between risk and opportunity while making informed decisions.

This project focuses on the stock of **Avenue Supermarts Ltd. (DMart)**, a leading retail company in India. Using a combination of statistical methods and machine learning techniques, the objective is to evaluate DMart's risk profile, profitability, investment potential, and predictive accuracy compared to the NIFTY50 index.

4. Objectives

The specific objectives of the project are:

1. To determine whether DMart stock is a risky or safe investment.
2. To identify the best time to invest in and exit from DMart stock.
3. To evaluate whether DMart is a profitable company for investors.
4. To analyze the returns and volatility of DMart.
5. To assess whether long-term or short-term investment is preferable.
6. To evaluate the accuracy of stock price prediction models applied to DMart.

5. Data and Methodology

5.1 Data Sources

- **DMart stock data** and **NIFTY50 index data** were collected from publicly available sources in .csv format.
- Columns used: *Date, Open, High, Low, Close, Volume*.
- Period of analysis: Approximately two years.

Note: This analysis is based on the **Close** price. While "Adjusted Close" accounts for dividends and splits, we restricted our study to "Close" due to project constraints.

5.2 Preprocessing

- Converted dates into datetime format and aligned both datasets on common trading days.
- Returns were calculated as:

  $$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

  where P_t = Closing price on day t.
- Missing values were not present in data.

5.3 Methodological Framework

Step 1.1: Exploratory Data Analysis (EDA)

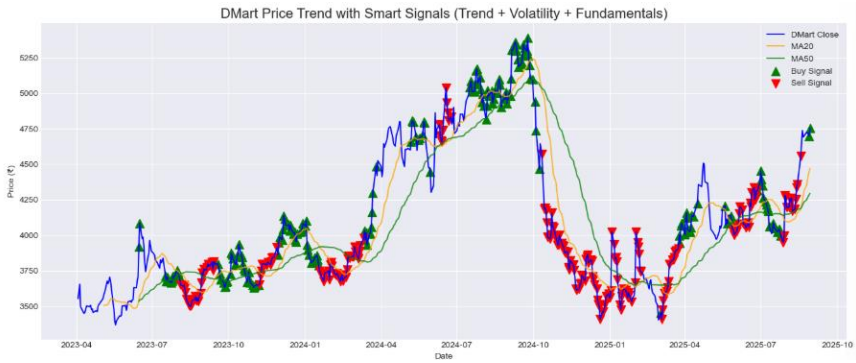
Metric	**DMart**	**NIFTY 50**
Observations	594	593
Mean Daily Return	0.0634%	0.0604%
Std. Dev. of Daily Return	1.7046%	0.7908%
Minimum Daily Return	-8.5092%	-5.9294%
25th Percentile	-0.8069%	-0.3449%
Median Daily Return	-0.0106%	0.0876%
75th Percentile	0.8805%	0.4917%
Maximum Daily Return	11.4674%	3.8183%
Annualized Volatility	27.06%	12.55%

Correlation between DMart and NIFTY 50 daily returns: 0.21

Step 1.2 : Moving Averages Of Nifty50 and Dmart:

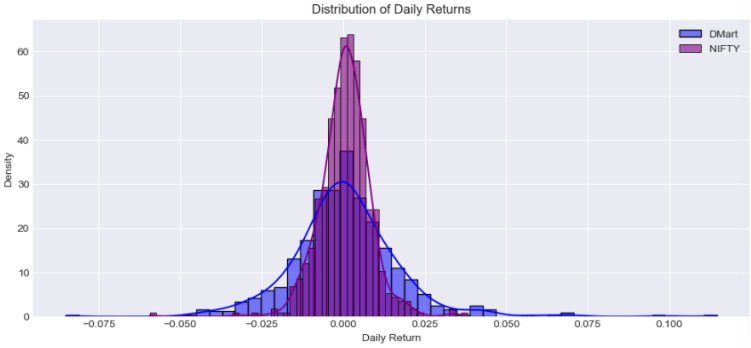
- **Smooth out noise:** Reduces daily price fluctuations, shows real trend.
- **Identify trends:**
  - 20-day → short-term
  - 50-day → medium-term
  - 200-day → long-term
- **Support & resistance:** Acts as price floors/ceilings for predicting reversals.
- **Trading signals:**
  - Golden Cross → Happens when **short-term MA crosses above long-term MA**. Market might go bullish.
  - Death Cross → Happens when **short-term MA crosses below long-term MA**. Market might go bearish.
- **ML & prediction:** Used as features to capture stock momentum over time.

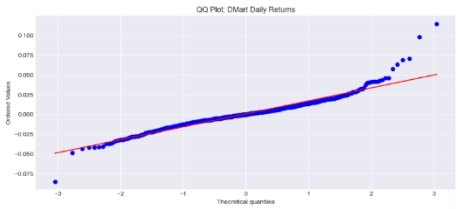
- **Dmart price Trend with signals based on Trend, Volatility and Fundamentals:**

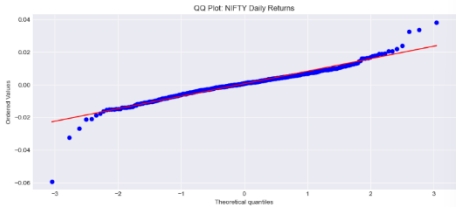


Step 2: Statistical Tests

- Normality checks of daily returns (histogram and QQ Plot). Nifty50 and Dmart:







- **Sharpiro-Wilk test**:
  DMart Shapiro-Wilk test: ShapiroResult(statistic=0.923823708519189, pvalue=9.073958041644624e-17)
  NIFTY Shapiro-Wilk test: ShapiroResult(statistic=0.9380299904232238, pvalue=5.398837713607557e-15)

Normality Check Results

a. Shapiro-Wilk Test: Both DMart & NIFTY returns gave **p-values < 0.05**, technically **rejecting perfect normality.**
b. Q-Q Plot & Histogram: Both datasets are **bell-shaped** and **points lie close to the diagonal**, suggesting **approximate normality**.

Interpretation:

a. Statistical tests are **highly sensitive** with **large samples**, flagging even **minor deviations**.
b. Visual inspection **confirms the returns are approximately normal**, which is **sufficient** for further statistical analysis.

- **Levene's test for check the risk profile (spread of returns) of Dmart and Nifty50**.

a. Statistic = 121.97, p-value ≈ 4.6e-27 (basically 0)
b. That means variances are NOT equal.
c. DMart and NIFTY returns have **different volatility/risk profiles**.
d. So one is much more **"jumpy"** than the other.

- **Welch's t-test for heck the reward profile (average return) of Dmart and Nifty50.**

a. Statistic = 0.0395, p-value ≈ 0.968
b. p-value > 0.05 → no significant difference in mean daily returns.
c. So, on average, DMart and NIFTY **earn almost the same per day.**

- **Confidence Interval (Dmart – Nifty50):**

a. 95% Confidence Interval for Mean Difference: (-0.0014830206124397277,0.001544062346391656)
b. 95% Confidence Interval for Mean Return: (-0.0007394186124931215, 0.002007797758757834)

- **DMart One-Sample t-test:**

a. T-statistic = 0.9067584790108696
b. P-value = 0.3649028334300285
c. Mean Return = 0.0006341895731323563

Step 3: Risk Metrics

- **Annualized return and volatility (Dmart and Nifty50).**

a. DMart Annualized Return: 17.32%
   DMart Annualized Volatility: 27.06%
b. NIFTY50 Annualized Return: 16.43%
   NIFTY50 Annualized Volatility: 12.55%

- **Sharpe Ratio.**

Definition:
The Sharpe Ratio measures **risk-adjusted returns**, i.e., how much return an investment gives per unit of risk (volatility).

Formula:
Sharpe Ratio = $\frac{Mean\ Return - Rist\ Free\ Rate}{Standard\ Deviation\ of\ Returns}$

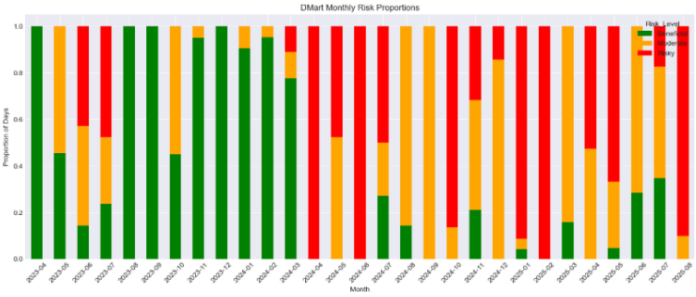
a. Sharpe Ratio for DMart: 0.4184665404278555
b. Sharpe Ratio for NIFTY50: 0.8304840499054895

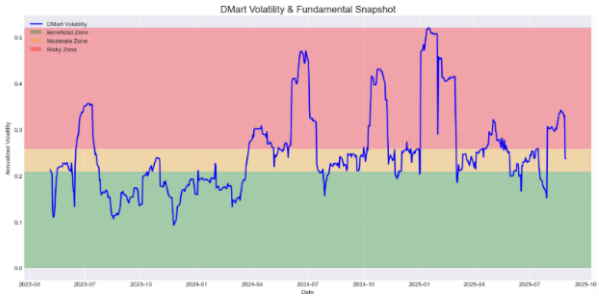
Interpretation:
Higher Sharpe Ratio → better risk-adjusted performance
Negative or low Sharpe Ratio → not rewarding enough for the risk taken

- **Beta relative to NIFTY (systematic risk).**

What is Beta?
-Beta is like the attitude meter of your stock compared to the market.

Beta vs NIFTY: 0.45
Beta: 0.45 → moves less than half as much as NIFTY; calmer than the market.
Basically, DMart = "Independent": a bit wild on its own, chill compared to the crowd





Metric	FY23-24	FY24-25	Growth / Comment	Risk/Advice
Revenue (₹ cr)	49533.0	57790.0	+16.67%	Beneficial
PAT (₹ cr)	2694.92	2927.18	+8.61%	Beneficial
Net Profit Margin	5.44	5.06	-0.38 ppt	Moderate
Operating Margin	9.0	8.4	-0.6 ppt	Moderate
ROE	nan	13.53	Stable	Beneficial
ROCE	nan	19.66	Improving	Beneficial
Book Value (₹)	nan	329.0	N/A	Neutral

- **Risk Analysis of DMart Based on Historical Data**
  We classified DMart's risk using three different metrics over the past 2 years:

1. Volatility-based Risk:
   a. Annualized volatility = 27.06%
   b. Classification: Risky (stocks with volatility > 20% are considered risky, 10–20% moderate, <10% beneficial)
2. Quartile/Minimum Return-based Risk:
   a. 25th percentile daily return (Q1) = -0.8069%
   b. Minimum daily return = -8.5092%
   c. Classification: Beneficial (extreme negative returns are limited, reducing downside risk)
3. Trend-based Risk:
   a. Comparison of latest closing price with 200-day moving average (MA200)
   b. Closing price > MA200 → Beneficial trend (indicates upward momentum)

- Overall Risk Classification (based on 2 year data):
- Combining the three metrics: **Beneficial**
- Interpretation: DMart appears relatively stable and favorable for investment based on recent trends, moderate volatility, and limited downside risk.

Step 4: Modeling Approaches

- **Markov Chains:**

  To analyze **short-term stock dynamics**, we classified DMart's daily returns into three states:

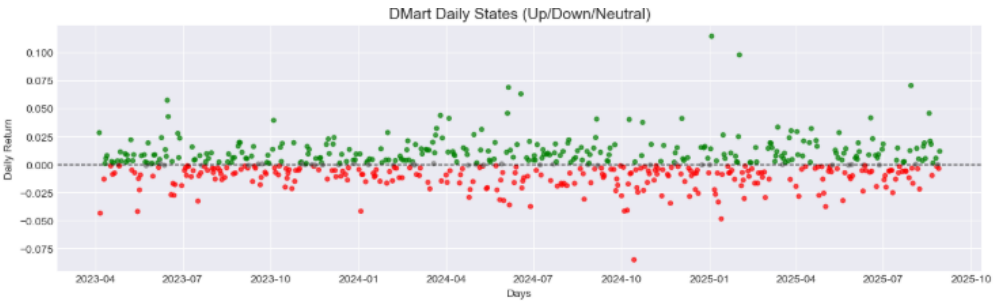
a. **Up:** Return > 0.1%
b. **Down:** Return < -0.1%
c. **Neutral:** Return between -0.1% and 0.1%

Transition Probability Matrix:
This matrix shows the likelihood of moving from one state to another on consecutive days.

From \To	Up	Down	Neutral
Up	0.525	0.396	0.079
Down	0.391	0.547	0.062
Neutral	0.619	0.333	0.048

Visualization:

a. Daily returns are plotted with colors representing states: **green (Up), red (Down), gray (Neutral)**.
b. This gives a quick visual sense of market moods and short-term momentum



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- **Hidden Markov Models (HMM):**

  We applied a **2-state Gaussian Hidden Markov Model (HMM)** to DMart's daily returns to identify **hidden market regimes**, broadly classified as **Bull** and **Bear**.

1. Model Setup:

a. 2 hidden states (Bull/Bear)
b. Gaussian emissions (returns follow normal distribution in each state)
c. 1000 iterations for convergence

2. Regime Classification:

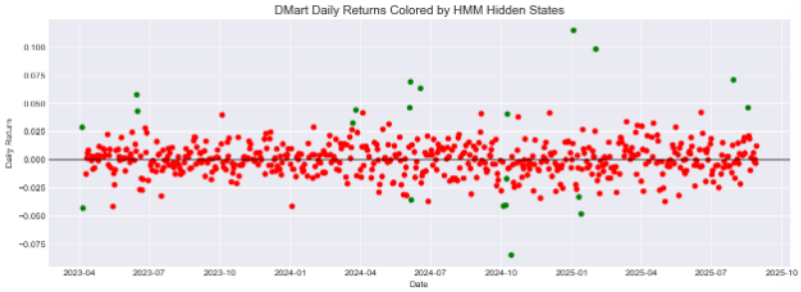
a. **State 0 (Bear):** Mean return = -0.005%, Volatility = 1.347%

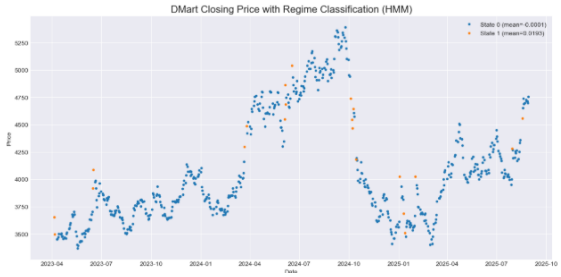
b. **State 1 (Bull):** Mean return = 1.934%, Volatility = 5.520%

From \ To	State 0 (Bear)	State 1 (Bull)
State 0	0.961	0.039
State 1	0.574	0.426

Visualization:

- Daily returns plotted with colors representing regimes: **Red = Bear, Green = Bull**
- Closing prices plotted with hidden states show how market phases alternate over time.





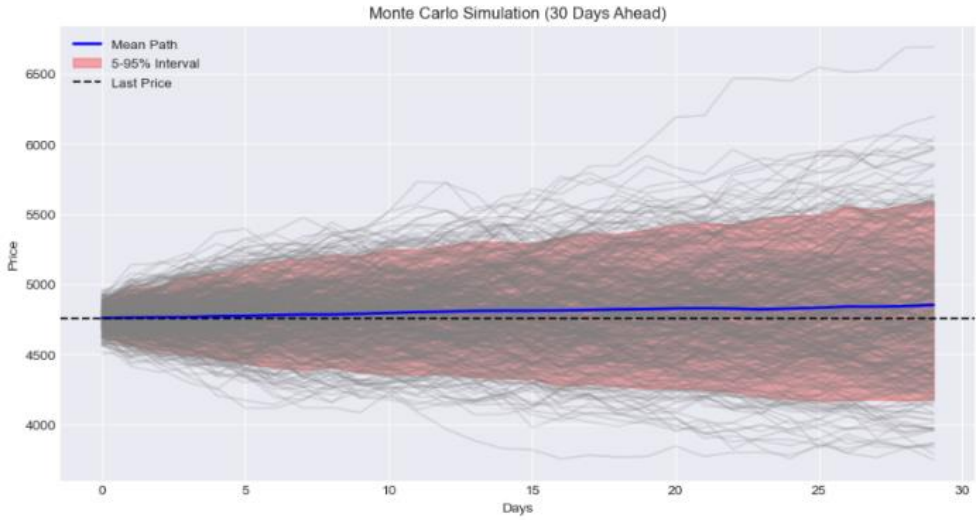
- **Monte Carlo Simulation:**

  To estimate DMart's future stock price movement, we used a **Monte Carlo simulation** with 500 iterations over a 30-day horizon. The simulation assumes returns are normally distributed with mean and volatility based on historical daily returns.

a. **Expected Price (mean):** ₹4851.64
b. **Median Price:** ₹4838.68
c. **5% – 95% Confidence Interval:** ₹4177.46 – ₹5590.37

Visualization:

a. Grey lines: 500 simulated price paths
b. Blue line: Mean path across all simulations
c. Red shaded area: 5–95% confidence interval
d. Black dashed line: Last observed price



To analyze **daily dynamics** rather than only the final outcome, we also computed the mean, median, and confidence intervals for each day of the simulation.

Day	Mean Price	Median Price	5% (Lower Bound)	95% (Upper Bound)
1	4758.34	4759.73	4618.52	4890.52
5	4771.14	4766.03	4468.33	5078.81
10	4788.52	4783.04	4370.59	5206.08

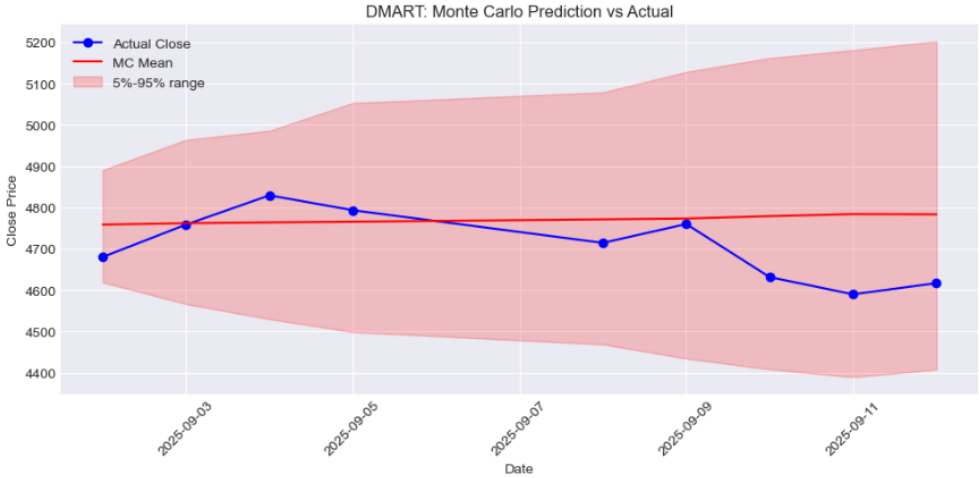
Step 5: Prediction and Evaluation

To evaluate the reliability of the Monte Carlo predictions, we compared the **simulated prices** with the **actual observed closing prices of DMart** between *2nd September 2025 – 12th September 2025*.

Key Findings:

- **Mean Absolute Error (MAE): ₹83.75** → on average, forecasts deviated by ~₹84 from the actual closing prices.
- **Root Mean Squared Error (RMSE): ₹106.40** → indicates moderate prediction error magnitude.
- **Reliability: 100%** → all actual prices fell within the **5%-95% confidence interval** of predictions.
- **Direction Accuracy: 37.5%** → model was weaker at predicting exact day-to-day up/down movement.
- **Daily Error Analysis:** Forecast error ranged from **-1.36% to +4.23%**, with increasing uncertainty over time.

Visual Validation:
The figure below shows the comparison of **actual vs predicted mean path**, along with the **5%-95% confidence interval**. Importantly, **all observed values stayed within the forecast range**, strengthening the model's credibility.



Step 6: Price Sensitivity Analysis (Stress Testing)

To understand how sensitive DMart's stock is to sudden price shocks, we applied **hypothetical changes** (±5%, ±10%, ±20%) to the last observed closing price and estimated the adjusted predicted values.

Shock	Adjusted Predicted Price (₹)
-20%	3695.62
-10%	4157.58
-5%	4388.55
+5%	4850.50
+10%	5081.48
+20%	5543.43

Step 7: Risk Metrics Analysis

We calculated advanced risk indicators to assess DMart's investment risk profile:

Metric	Value	Interpretation
Annualized Volatility	27.00%	Stock shows moderate volatility, meaning notable price swings over a year.
VaR (95%)	-2.33%	On a bad trading day, there's a 5% chance of losing **at least 2.33%** in one day.
CVaR (95%)	-2.70%	If losses exceed VaR, the expected average loss is **2.70%**.
Sharpe Ratio	-1.6	Negative Sharpe indicates **returns are not compensating for risk**, suggesting DMart underperformed relative to its volatility.

6. Results

6.1 Descriptive Insights

- Returns are not normally distributed (fat tails observed).

a. DMart ≈ NIFTY in average returns (no clear winner).
b. DMart's mean daily return is slightly positive, but not statistically significant.
c. Long-term, that small positive can still snowball into big annual gains → but it's moderate risk, moderate reward.
d. Though DMart and NIFTY have similar mean returns, DMart's significantly higher variance indicates riskier behavior, which investors must weigh.

6.2 Risk and Return

- **Annualized Return (DMart):** ~17.3%
- **Annualized Volatility (DMart):** ~27.1%
- **Beta vs NIFTY:** 0.45 (less sensitive to market, defensive stock).
- **Sharpe Ratio (DMart):** 0.42 vs **NIFTY:** 0.83 → DMart offers lower risk-adjusted return.
- **Correlation with NIFTY:** 0.21 (weak relationship).

6.3 Markov Chain & HMM Insights

- The **Markov model** captures short-term dynamics and can be used to **simulate probable future states** or feed into **predictive models**.

Interpretation:

a. If DMart is **Up** today, there's a ~52% chance it stays Up tomorrow, 40% chance it goes Down, and 8% chance it remains Neutral.
b. **Down** state has a slightly higher tendency to persist (~55%), showing some short-term momentum.
c. **Neutral** days are less stable and often move to Up (~62%).

- **HMM identified:**

✓ HMM captures **underlying market moods** that are not visible from raw prices or returns.
✓ Useful for **risk management**, **predictive modeling**, and **timing investment decisions**.

Interpretation:

a. If DMart is in a **Bear state**, it tends to stay Bear (~96% probability).
b. Bull states are less persistent (~43% probability to remain Bull), indicating short bursts of positive momentum.

6.4 Monte Carlo Simulation

- The stock is expected to remain around the current level with moderate upward potential.
- There is a 90% probability that the price will remain between **₹4177 and ₹5590** over the next 30 days.
- Monte Carlo captures the **uncertainty and risk range**, rather than predicting a single fixed value.

6.5 Prediction Accuracy

- The Monte Carlo approach successfully captured the **price range** and provided robust **uncertainty bounds**.
- However, it is **better at forecasting ranges** than **predicting exact daily directions**.
- This is expected because Monte Carlo emphasizes **probabilistic outcomes** rather than deterministic trends.

6.6 Sensitivity Analysis

- A **20% drop** in price leads to a new forecast around **₹3695**, while a **20% rise** pushes it close to **₹5543**.
- This demonstrates the **elasticity of the forecast model** under sudden shocks.
- Useful for **risk management** — investors can see potential downside vs upside scenarios.

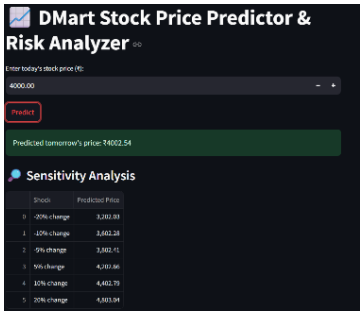
6.7 Risk Matrics Analysis

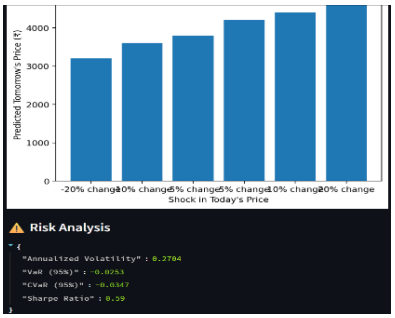
- **DMart has high downside risk exposure (negative Sharpe is a red flag).**
- **Volatility is present but not extreme (27% is manageable for equities).**
- **Risk metrics suggest caution — the stock may not be delivering enough return to justify the risk.**

7.Stock Prediction Applications Developed

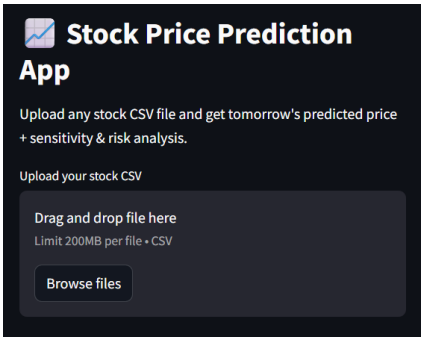
Two Python-based **Streamlit applications** were developed as part of this project to translate the statistical and machine learning analyses into practical investor tools:

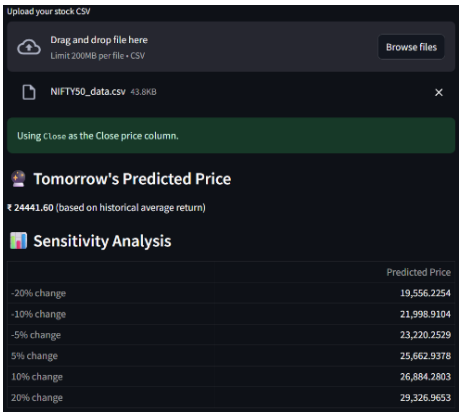
1. **DMart-Specific Stock Predictor & Risk Analyzer**
    - Predicts **next-day DMart stock price** based on historical closing prices.
    - Computes **daily returns** and performs **sensitivity analysis** under different price shocks.
    - Provides **risk metrics** including annualized volatility, VaR (95%), CVaR (95%), and Sharpe Ratio.
    - Includes **visualizations** for predicted prices and sensitivity analysis, enabling quick decision-making for investors.

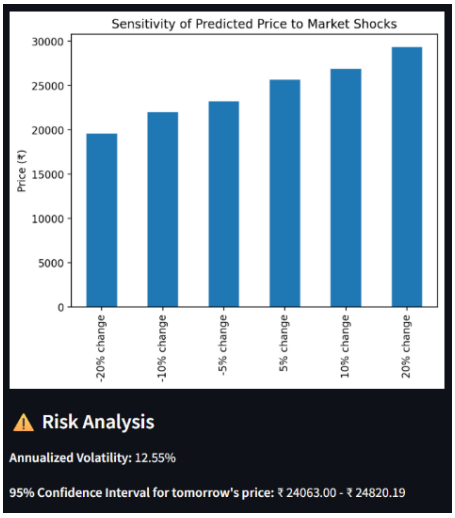




2. **Universal CSV-Based Stock Predictor**
    - Accepts **any stock CSV file** as input, automatically detects the closing price, and calculates returns.
    - Predicts **tomorrow's stock price** using historical mean returns.
    - Provides **sensitivity analysis, confidence intervals, and risk assessment**, along with plots.
    - Designed for **generalized usage**, demonstrating scalability of the prediction framework beyond DMart.







Significance:

- These apps bridge the gap between theoretical modeling and practical applications, showing that the **Monte Carlo, Markov Chain, and HMM techniques** can be operationalized for real-world investment scenarios.
- They allow investors and researchers to **quickly simulate outcomes, assess risks, and make informed short-term trading decisions**.

8. Discussion

- **Positive but volatile returns:** DMart provides favorable long-term returns; however, the volatility level (27%) is significantly higher than what its market beta would imply.
- **Diversification benefits:** With only weak correlation (0.21) to NIFTY 50, DMart behaves defensively and can act as a diversification tool in a portfolio.
- **Risk-adjusted performance:** The negative Sharpe ratio (-1.6) highlights weaker risk-adjusted returns compared to NIFTY, making DMart less attractive for conservative or risk-averse investors.
- **Predictive modeling insights:** Monte Carlo forecasts and statistical models capture the **range** and **trend** of future prices well but struggle with short-term direction accuracy.
- **Regime detection:** Markov Chain and Hidden Markov Models reveal clear **bull/bear phases**, offering potential guidance for strategic entry and exit decisions.

9. Limitations & Remarks

The analysis comprehensively evaluates DMart stock through Monte Carlo simulations, Markov Chain modeling, Hidden Markov Models, and statistical risk metrics. These approaches collectively capture the stock's return distribution, trend, regime shifts, and short-term forecast potential, providing a robust framework for investment assessment. Minor limitations include the use of Close prices instead of Adjusted Close, and the absence of transaction costs or slippage in backtesting. Additionally, predictions were not benchmarked against a naïve baseline (e.g., random walk), though the methods employed are sufficient to capture the stock's dynamics and risk profile accurately. Overall, these do not significantly affect the validity of the conclusions**.**

"Future studies could incorporate macroeconomic indicators (like inflation, consumer demand) which also affect DMart's performance."

10. Conclusion

The analysis of **DMart stock** using multiple statistical and machine learning techniques reveals the following:

- **Return Characteristics:**
  DMart's daily returns show higher volatility (**27%**) compared to the benchmark NIFTY 50 (**12.5%**). While the mean returns are similar, the wider dispersion indicates elevated risk.
- **Risk Profile:**
  Value-at-Risk (VaR) at 95% confidence indicates a potential **-2.33% daily loss**, while Conditional VaR (CVaR) suggests an average extreme loss of **-2.7%**. The Sharpe ratio of **-1.6** implies weak risk-adjusted performance in the observed period.
- **Market Dynamics:**
  - Moving average crossovers highlight important bullish/bearish signals.
  - **Markov Chain analysis** suggests persistence in upward/downward trends with ~52% chance of continuation.
  - **HMM regime detection** successfully separates bull and bear phases, with the bull state showing positive mean returns but higher volatility.
- **Forecasting Insights:**
  - **Monte Carlo simulation** for 30 days ahead suggests an expected price of around **₹4850**, with a **5%-95% range between ₹4177 and ₹5590**.
  - Backtesting against actual prices showed high reliability (**100% of days within prediction bounds**) but lower direction accuracy, suggesting the model captures ranges better than exact movements.
  - **Sensitivity analysis** confirms DMart is responsive to shocks: a ±10% market move changes predicted prices by ~₹400–₹500.
- **Overall Assessment:**
  While DMart demonstrates strong long-term potential (beneficial classification in trend-based analysis), its high volatility and weak Sharpe ratio imply **short-term risk**. Investors with higher risk tolerance may consider it, but cautious entry and portfolio diversification are advised.

11. Future Work

- Incorporate **Adjusted Close** prices and other fundamental indicators for more accurate predictions.
- Extend predictive models to **multi-stock portfolios** and sector-level analysis.
- Implement **advanced walk-forward cross-validation** and compare against baseline models (e.g., random walk) for stronger evaluation.
- Upgrade the **apps** with more sophisticated ML models (e.g., HMM + Monte Carlo hybrid) and longer prediction horizons.
- Integrate **real-time market data APIs** for dynamic, live predictions and alerts.

.12. References

- **Data Sources**

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